Unit: Unit 0: Foundations of Algebra
Topic: Natural Numbers Integers

Name: ____________
Date: ________

Foundation (Jalapeno)

Q1. Classify the number 5 as a:
(A) natural and whole number
(B) natural number only
(C) whole number only
(D) integer only
(E) none of the above

Q2. Which of the following is an example of an odd number in the context of planetary orbits?
(A) the 2^{nd} planet from the Sun
(B) the 3^{rd} planet from the Sun
(C) the 5^{th} planet from the Sun
(D) the 7^{th} planet from the Sun
(E) the 9^{th} planet from the Sun

Q3. Identify the next number in the sequence: $2, 5, 8, 11, 14, ...$
(A) 15
(B) 17
(C) 18
(D) 20
(E) 22

Q4. Evaluate the expression $(-3) + (-2)$
(A) -1
(B) -5
(C) -6
(D) 1
(E) 5

Q5. What is the result of $4 \cdot (-9)$?
(A) -36
(B) 36
(C) -4
(D) 9
(E) 0

Q6. Which of the following numbers is even: 23, 37, 48, 52, 67?
(A) 23
(B) 37
(C) 48
(D) 52
(E) 67

Q7. Solve for x: $x - 2 = 7$
(A) 5
(B) 7
(C) 9
(D) 10
(E) 12

Q8. Classify the number -11 as a:
(A) natural number
(B) whole number
(C) integer
(D) rational number
(E) irrational number

Open Ended Questions (FRQ)

Q9. A spacecraft is traveling at a speed of -25 km/h. If it continues at this speed, how far will it have traveled after 4 hours? Show your work and explain your answer.

Q10. Create a sequence of 5 consecutive odd integers, starting from -3. Graph the sequence on a number line and identify any patterns you observe. Explain the reasoning behind your sequence.

Chef's Tips

- Tip 1: Encourage students to read each question carefully and choose the correct answer.
- Tip 2: Remind students to show their work for the free-response questions.
- Tip 3: Allow students to use a calculator for the free-response questions if necessary.